

HELLIGVAER, Norway

WMO: 011440

Lat: 67.405N

Lon: 13.896E

Elev: 79

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.1	21.9	7.6	8.2	21.1	10.2	9.3	23.7	49.8	36.2	44.6	34.7	23.1	110	0.898

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	67.8	59.2	64.7	57.4	62.2	56.1	60.1	66.0	58.4	63.1	57.0	61.1	10.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	70.5	61.9	56.1	67.1	60.6	54.7	63.9	59.3	26.5	66.1	25.4	63.2	24.5	61.3	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
41.9	36.8	32.9	DB	17.3	73.0	3.8	4.4	14.6	76.2	12.4	78.7	10.2	81.2	7.5	84.3
			WB	14.8	62.8	3.4	2.7	12.3	64.7	10.3	66.3	8.3	67.8	5.8	69.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	33.0	32.7	33.8	39.0	45.2	49.9	55.5	55.5	51.3	44.5	39.4	36.2	
	DBStd	9.50	5.63	5.62	5.00	4.33	4.82	4.68	4.79	3.90	4.01	4.79	5.12	5.11	
	HDD50	3025	526	485	503	331	165	58	2	2	31	177	317	428	
	HDD65	8011	991	905	968	779	615	454	299	295	410	635	767	893	
	CDD50	495	0	0	0	1	15	54	173	173	71	8	0	0	
	CDD65	5	0	0	0	0	0	0	4	1	0	0	0	0	
	CDH74	12	0	0	0	0	0	0	11	1	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	16.8	21.3	20.7	18.7	15.4	14.0	13.3	12.3	12.2	15.0	18.3	19.6	21.5	
Precipitation	PrecAvg	66.6	7.3	6.5	6.2	3.9	3.1	3.6	4.0	4.4	6.3	7.0	6.1	8.1	
	PrecMax	83.0	15.4	13.7	11.2	9.1	4.6	7.1	9.1	8.2	13.7	13.7	12.6	13.5	
	PrecMin	52.1	0.6	1.2	1.9	1.4	1.3	1.0	1.7	1.0	2.2	2.4	0.7	2.8	
	PrecStd	8.2	3.7	3.3	2.8	1.9	0.9	1.5	1.7	1.7	3.1	2.7	3.2	3.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	43.9	46.1	55.6	63.9	67.5	75.8	69.3	63.4	56.8	51.3	47.2	
		MCWB	41.7	40.9	40.7	46.3	52.2	56.0	63.0	59.8	55.6	50.1	48.6	43.8	
	2%	DB	43.7	42.6	43.5	50.4	58.6	63.1	69.6	66.2	60.3	54.4	49.1	45.7	
		MCWB	41.0	40.2	40.1	43.5	50.6	54.5	61.1	58.4	53.6	49.2	46.0	43.7	
	5%	DB	42.1	41.7	42.2	46.7	54.9	59.9	65.8	63.7	58.5	52.6	47.5	44.5	
		MCWB	39.4	39.3	39.4	41.7	48.2	53.3	58.7	57.4	52.9	48.7	44.4	42.3	
	10%	DB	40.7	40.6	40.9	44.3	51.8	56.9	62.8	61.6	56.8	50.9	46.1	43.2	
		MCWB	38.1	38.1	38.1	40.6	45.9	51.7	57.1	56.3	52.2	47.7	43.0	40.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	42.7	42.3	46.6	53.4	57.5	64.0	61.5	57.1	52.0	49.3	45.8
			MCDB	44.0	43.0	43.4	54.3	62.0	64.9	72.3	67.2	60.3	54.5	50.4	46.1
2%		WB	41.2	41.1	41.3	44.3	51.1	55.5	61.4	59.5	55.3	50.5	46.7	44.4	
		MCDB	42.7	42.1	43.0	48.6	57.8	61.6	68.8	64.3	58.3	53.2	48.5	45.3	
5%		WB	39.7	39.7	39.8	42.9	48.7	53.8	59.3	58.2	54.1	49.3	45.0	42.7	
		MCDB	41.5	41.3	42.0	45.5	54.1	58.7	64.6	62.6	57.1	52.0	47.1	44.4	
10%		WB	37.9	38.1	38.3	41.6	46.8	52.2	57.6	56.9	53.0	48.0	43.4	40.6	
		MCDB	40.4	40.3	40.8	44.0	50.8	55.8	61.8	60.7	56.0	50.4	45.9	42.9	
Mean Daily Temperature Range		5% DB	MDBR	5.2	5.4	5.4	6.1	7.0	6.2	7.0	6.8	5.6	4.7	4.8	5.0
			MCDBR	6.2	6.3	6.3	9.2	11.7	11.3	12.4	11.1	7.9	5.8	5.2	6.4
	MCWBR		6.3	6.6	5.7	5.2	6.1	6.0	6.3	5.2	4.4	4.7	5.3	6.7	
	5% WB	MCDBR	6.4	6.7	6.5	7.3	10.9	9.8	11.5	9.7	7.0	5.2	5.4	6.5	
		MCWBR	6.8	7.3	6.3	5.3	6.3	5.7	6.3	5.4	4.7	5.1	5.9	7.1	
Clear-Sky Solar Irradiance	taub	0.365	0.292	0.322	0.341	0.345	0.342	0.349	0.339	0.326	0.297	0.348	N/A		
	taud	2.574	2.186	2.299	2.392	2.460	2.510	2.518	2.534	2.534	2.303	2.529	N/A		
	Ebn at Noon	48	165	228	259	271	274	268	259	233	163	42	0		
	Edh at Noon	5	15	23	28	29	29	28	24	19	13	4	0		
All-Sky Solar Radiation	RadAvg	12	112	459	997	1456	1628	1472	1092	580	200	30	0		
	RadStd	1	18	61	97	104	213	154	107	59	39	3	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page